Exp No: 14:

Determination and plotting of Polarization in different Dielectric Medium with their relative permittivity and to find electric susceptibility.

Date:

Aim: To determine and plot polarization in different dielectric medium with their relative permittivity and to find electric susceptibility.

Software required:

- SCILAB version 6.0.1

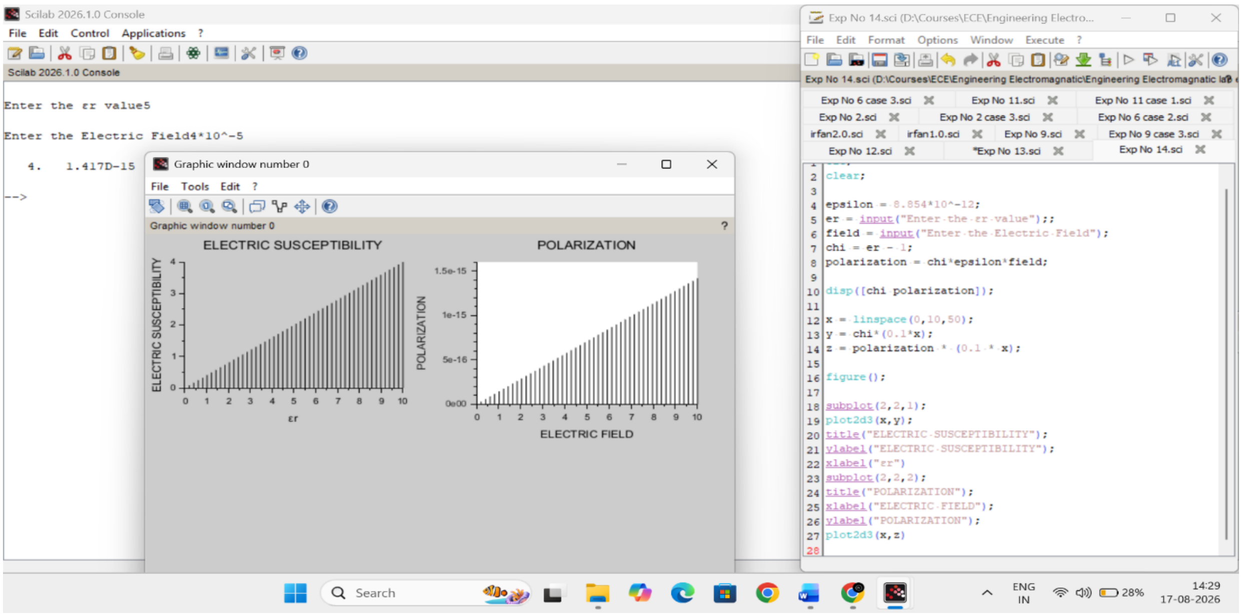
Theory:

In a dielectric material, when an electric field is applied, the molecules or atoms within the material become polarized. This means that the positive and negative charges within the atoms or molecules shift slightly, creating an internal electric dipole moment. The **polarization** P is defined as the dipole moment per unit volume of the material and is given by the equation:

$$\mathbf{P} = \epsilon_0 \chi_e \mathbf{E}$$

Where:

- $\mathbf{P}$ is the polarization vector (Coulombs per square meter),
- ϵ_0 is the permittivity of free space (approximately $8.854 \times 10^{-12}\ \mathrm{F/m}$),
- χ_e is the electric susceptibility of the material,
- $\mathbf{E}$ is the applied electric field (Volts per meter).



Exp No :- 14

Finding electric susceptibility
Polarization

$\varepsilon_r = 5$
$E = 4 \times 10^{-5}$

Electric susceptibility

$$\chi_e = \varepsilon_r - 1$$
$$= 5 - 1$$

$$\boxed{\chi_e = 4}$$

Polarization

$$P = \chi_e \varepsilon_0 E$$

$$= (4)(8.854 \times 10^{-12})(4 \times 10^{-5})$$

$$= 4 \times 8.854 \times 10^{-12}$$

$$= 3.5416 \times 10^{-11} (4 \times 10^{-5})$$

$$P = 1.4166 \times 10^{-15}$$

$$\boxed{P = 1.417 \times 10^{-15}}$$

Result:

Thus, the determination and plotting of polarization in different dielectric medium with their relative permittivity and to find electric susceptibility is executed.

Case 1: To determine and plot the polarization and electric susceptibility of electromagnetic waves in three dielectric media: Air, Water, and Glass, using their relative permittivity.

Analytical:
Determine and plot the polarization and electric susceptibility of electromagnetic waves in three dielectric media: Air, Water, and Glass, using their relative permittivity.

- **Air:**
  - Relative Permittivity: $\epsilon r=1$
  - Electric Susceptibility: χe (to be calculated)
- **Water:**
  - Relative Permittivity: $\epsilon r=80$
  - Electric Susceptibility: χe
- **Glass (e.g., Borosilicate Glass):**
  - Relative Permittivity: $\epsilon r=6$
  - Electric Susceptibility: χe

Solution:

1. Air:

- Relative Permittivity $\varepsilon_r = 1$
- Electric Susceptibility $\chi_e = \varepsilon_r - 1 = 1 - 1 = 0$

Since the electric susceptibility of air is zero, there is no polarization in air under an applied electric field. The polarization $\mathbf{P}$ is also zero.

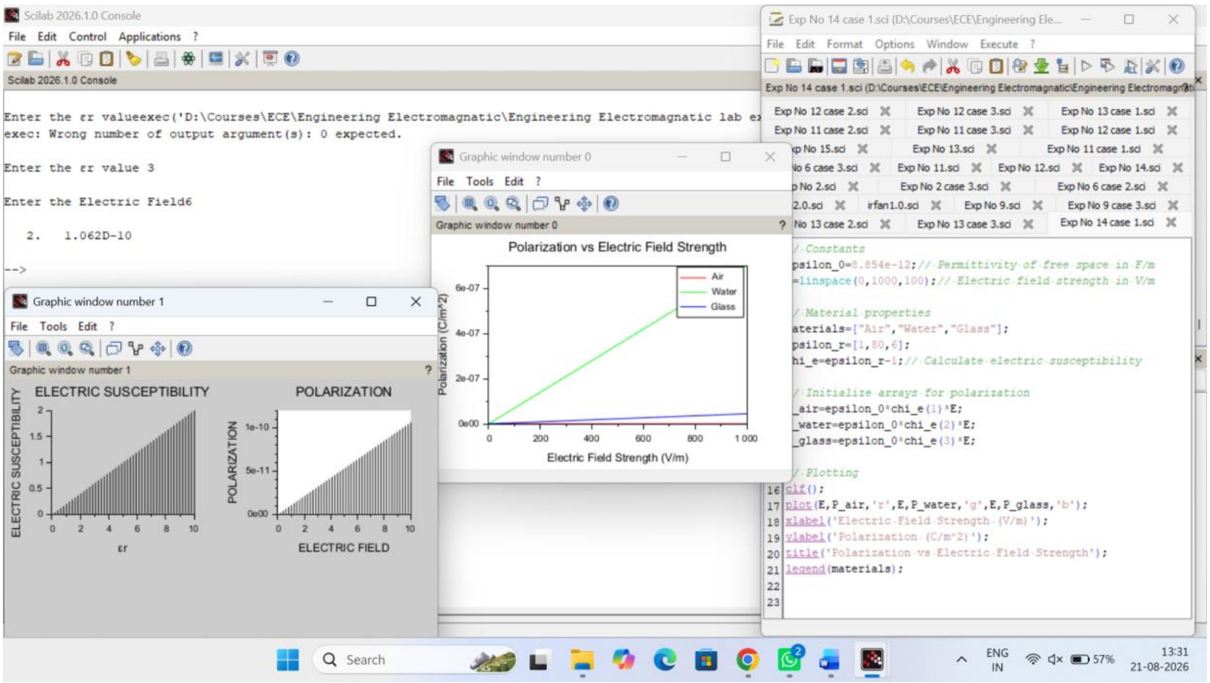
2. Water:

- Relative Permittivity $\varepsilon_r = 80$
- Electric Susceptibility $\chi_e = \varepsilon_r - 1 = 80 - 1 = 79$

Water has a high electric susceptibility, which means it will exhibit significant polarization under an applied electric field. The polarization is proportional to the electric field $\mathbf{P} = \epsilon_0 \chi_e \mathbf{E}$, where $\chi_e = 79$.

3. Glass (Borosilicate Glass):

- Relative Permittivity $\varepsilon_r = 6$
- Electric Susceptibility $\chi_e = \varepsilon_r - 1 = 6 - 1 = 5$



Result:

Thus, the determination and plotting of polarization in different dielectric medium with their relative permittivity and to find electric susceptibility is executed.

Case 2: To calculate and plot the polarization and electric susceptibility in Silicon, Teflon, and Paper, and understand how their relative permittivity affects the polarization.

Analytical:

To calculate and plot the polarization and electric susceptibility in Silicon, Teflon, and Paper

- **Silicon:**
  - Relative Permittivity: $\epsilon r=11.7$
  - Electric Susceptibility: χe
- **Teflon:**
  - Relative Permittivity: $\epsilon r=2.1$
  - Electric Susceptibility: χe
- **Paper:**
  - Relative Permittivity: $\epsilon r=3.0$
  - Electric Susceptibility: χe

Solution:

1. **Silicon:**

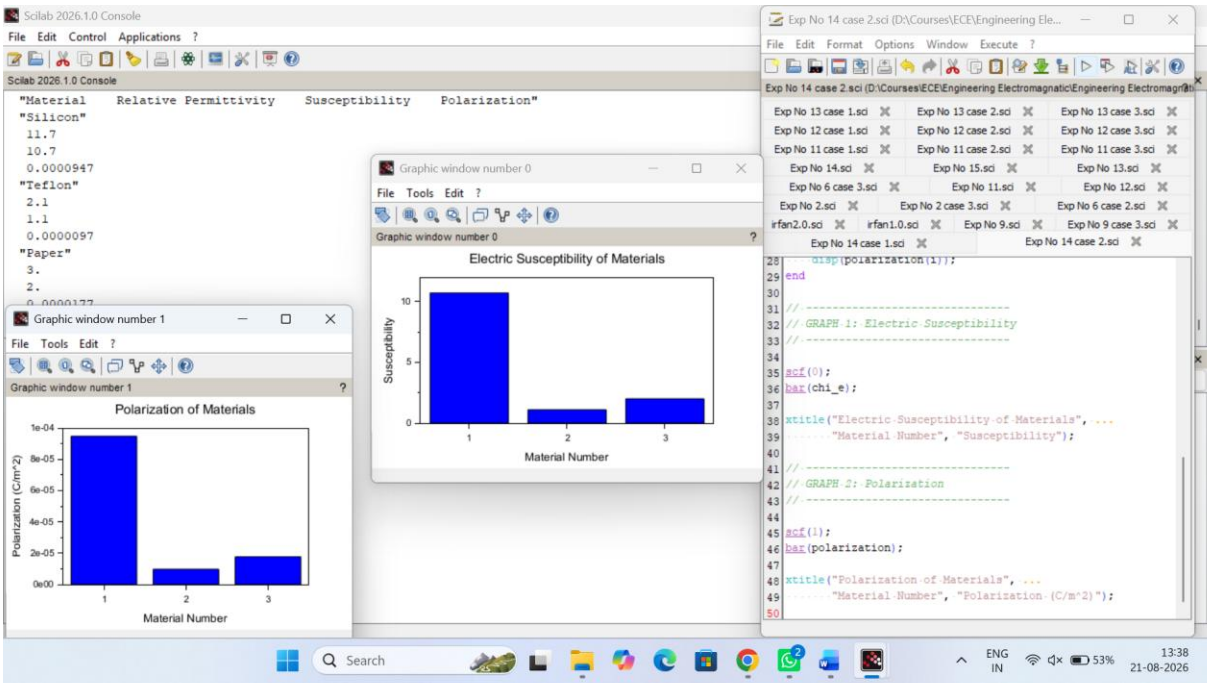
- Relative Permittivity $\varepsilon_r = 11.7$
- Electric Susceptibility $\chi_e = \varepsilon_r - 1 = 11.7 - 1 = 10.7$

2. **Teflon:**

- Relative Permittivity $\varepsilon_r = 2.1$
- Electric Susceptibility $\chi_e = \varepsilon_r - 1 = 2.1 - 1 = 1.1$

3. **Paper:**

- Relative Permittivity $\varepsilon_r = 3.0$
- Electric Susceptibility $\chi_e = \varepsilon_r - 1 = 3.0 - 1 = 2.0$



Result:

Thus the determination and plotting of polarization in different dielectric medium with their relative permittivity and to find electric susceptibility is executed.

Case 3: To explore how different dielectric materials (Rubber, Oil, and Plastic) affect the polarization and electric susceptibility, and to compare their relative permittivity's.

Analytical:

Calculate how different dielectric materials (Rubber, Oil, and Plastic) affect the polarization and electric susceptibility, and to compare their relative permittivity's.

- **Rubber:**
  - Relative Permittivity: $\epsilon r=3.5$
  - Electric Susceptibility: χe

- **Oil:**
  - Relative Permittivity: ϵr=2.2
  - Electric Susceptibility: χe
- **Plastic (e.g., Polyethylene):**
  - Relative Permittivity: ϵr=2.3
  - Electric Susceptibility: χe

Solution:

1. Rubber:

- Relative Permittivity $\varepsilon_r = 3.5$
- Electric Susceptibility $\chi_e = \varepsilon_r - 1 = 3.5 - 1 = 2.5$

The **polarization** is given by:

$$\mathbf{P}_{\text{rubber}} = \epsilon_0 \times \chi_e \times \mathbf{E} = 8.854 \times 10^{-12} \times 2.5 \times 1 = 2.2135 \times 10^{-11}\ \mathrm{C/m^2}$$

2. Oil:

- Relative Permittivity $\varepsilon_r = 2.2$
- Electric Susceptibility $\chi_e = \varepsilon_r - 1 = 2.2 - 1 = 1.2$

The **polarization** is given by:

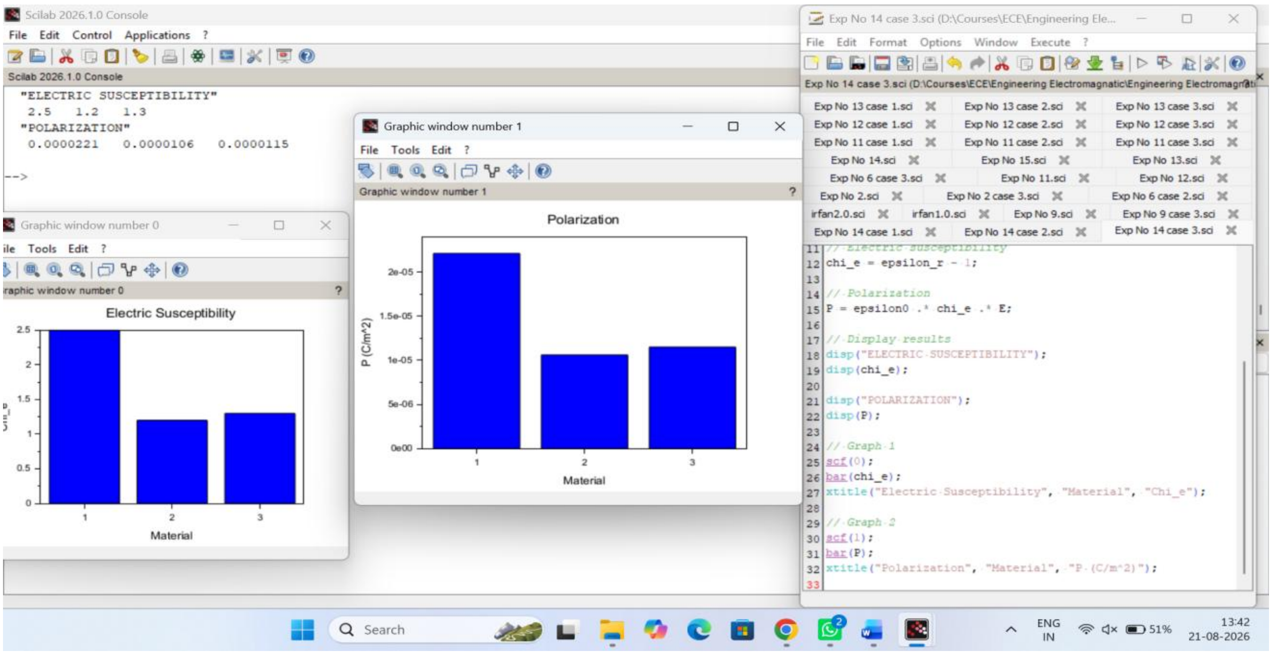
$$\mathbf{P}_{\text{oil}} = \epsilon_0 \times \chi_e \times \mathbf{E} = 8.854 \times 10^{-12} \times 1.2 \times 1 = 1.0625 \times 10^{-11}\ \mathrm{C/m^2}$$

3. Plastic (Polyethylene):

- Relative Permittivity $\varepsilon_r = 2.3$
- Electric Susceptibility $\chi_e = \varepsilon_r - 1 = 2.3 - 1 = 1.3$

The **polarization** is given by:

$$\mathbf{P}_{\text{plastic}} = \epsilon_0 \times \chi_e \times \mathbf{E} = 8.854 \times 10^{-12} \times 1.3 \times 1 = 1.151 \times 10^{-11}\ \mathrm{C/m^2}$$



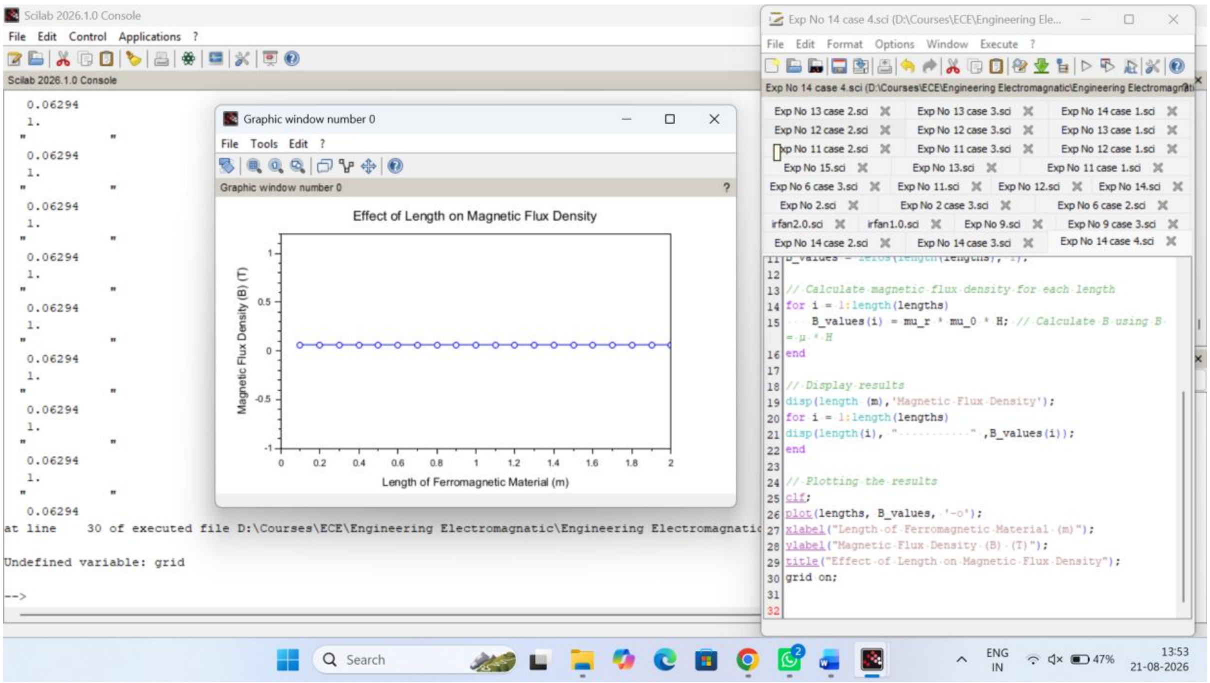
Result:

Parameters	Simulated Result	Theoretical result
εr value = 3.5 - Rubber	2.2135x 10-11	2.2135x 10-11
εr value = 2.2 – Oil	1.0625x10-11	1.0625x10-11
εr value = 2.3– Plastic	1.151x10-11	1.151x10-11

Thus, the determination and plotting of polarization in different

Case 1: Examine the effect of length of a ferromagnetic material on the magnetic flux density (B) for the following conditions

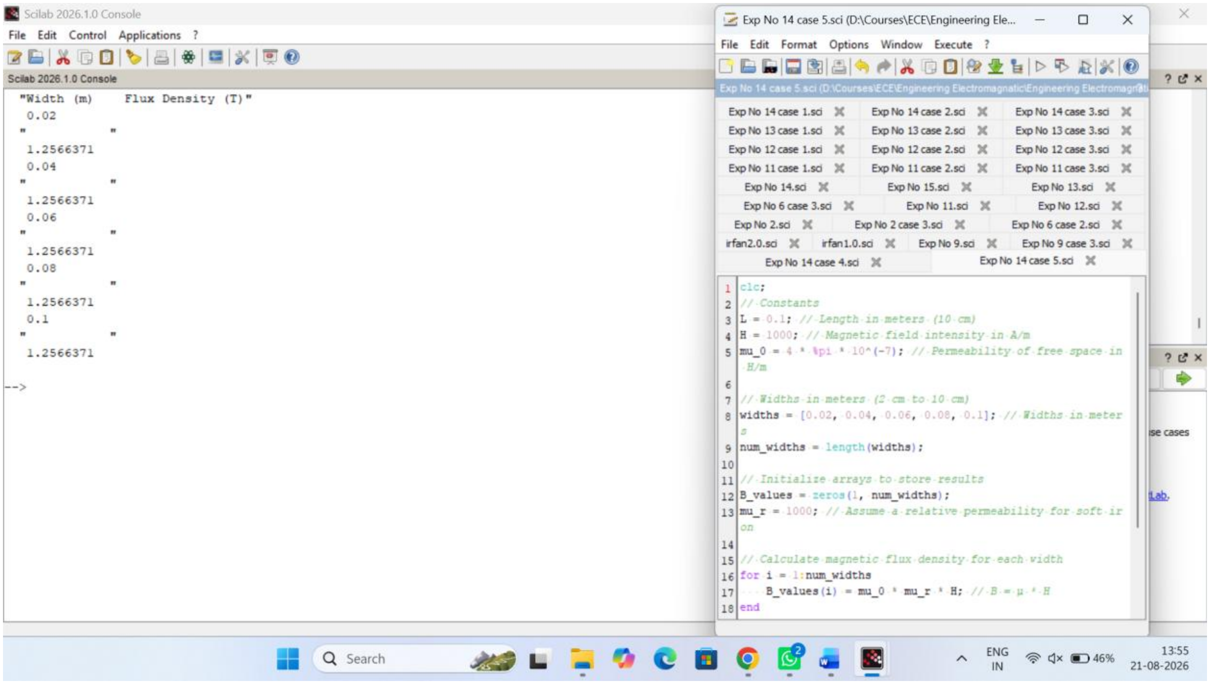
- The material is ferromagnetic with relative permeability (μr) of 500.
- The magnetic field intensity (H) is held constant at 100 A/m.
- The length of the material varies from 0.1 m to 2 m.



Case 2: To investigate how the magnetic flux density (B) in a ferromagnetic material changes with different widths at a constant length and magnetic field intensity.

Conditions:

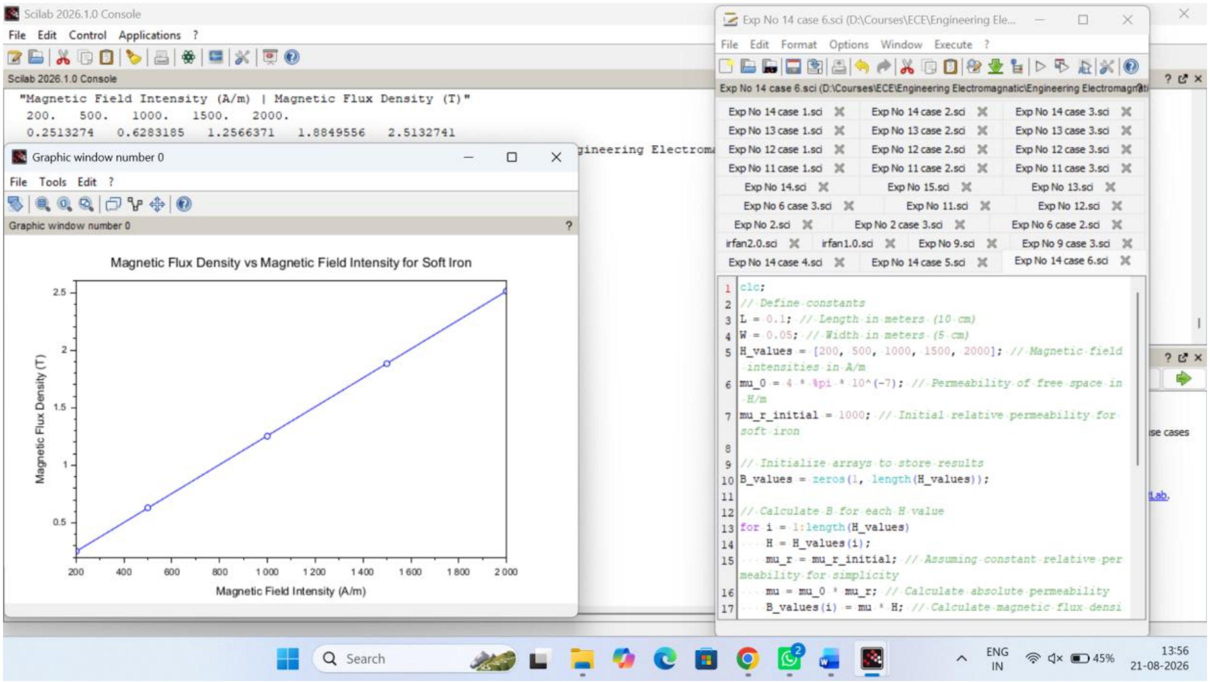
- **Material**: Soft iron (same as in Case Study 1)
- **Length (L)**: Constant at 10 cm
- **Magnetic Field Intensity (H)**: Constant at 1000 A/m
- **Width (W)**: Varies from 2 cm to 10 cm (i.e., 2 cm, 4 cm, 6 cm, 8 cm, 10 cm)



Case 3: To examine how the magnetic flux density (B) in a ferromagnetic material changes with different values of magnetic field intensity at constant length and width.

Conditions:

- Material: Soft iron (same as in the previous studies)
- Length (L): Constant at 10 cm
- Width (W): Constant at 5 cm
- Magnetic Field Intensity (H): Varies from 200 A/m to 2000 A/m (i.e., 200 A/m, 500 A/m, 1000 A/m, 1500 A/m, 2000 A/m)



Result:

Thus the program was verified for program for Magnetic flux density in ferromagnetic materials using SCILAB was successfully.